



# PortFlow AI

Autonomous Maritime Terminal Operations Intelligence Platform



## Predictive Congestion

6h to 72h continuous quayside occupancy forecasting with XGBoost models.



## OR-Tools Dispatch

Dynamic berth scheduling and crane gang allocation minimizing vessel dwell time.



## IBM Bob RAG Copilot

Grounded conversational operations advisor synthesizing live port telemetry into playbooks.

# The High Cost of Port Gridlock

Over 80% of world trade moves through marine terminals, yet operations remain bottlenecked by reactive legacy systems.

**\$50K+**

## Daily Demurrage / Ship

Anchored container ships waiting for berth access incur massive financial penalties daily.

**38%**

## Dwell Time Wastage

Unsynchronized crane gangs and tidal window conflicts paralyze quayside throughput.

**0h**

## Predictive Horizon

Legacy TOS (Navis N4, spreadsheets) report past logs instead of proactive 72h forecasting.

# Decoupled, Event-Driven Intelligence Layer

Fusing real-time AIS transponders, quayside telemetry, and IBM Bob AI into an interactive cockpit.



## Presentation & Digital Twin

- 60 FPS HTML5 Canvas maritime chart with vessel AIS vectors and berth HUD.
- 7 modular consoles: Command Center, Port Map, Predictions, Optimization, Route Advisor, Simulator, Copilot.



## AI & Optimization Engine

- Multi-horizon predictive forecasters for 6h, 12h, 24h, 48h, 72h windows.
- Google OR-Tools Mixed-Integer Linear Programming for berth/crane dispatch.
- IBM Bob RAG Engine grounding watsonx.ai in live terminal state vectors.

# Built for High-Stakes Harbor Operations

Empowering harbor masters and terminal directors to resolve crises before vessels cross the breakwater.

## Canvas GIS Digital Twin

Real-time AIS positioning, depth draft verification, and crane density telemetry.

## Disruption Sandbox

Simulate crane breakdowns and storms with instant before-and-after delta impact.

## IBM Bob Copilot

Natural-language crisis guidance with 1-click action execution for operators.

# Quantified Business & Environmental ROI

Direct operational savings for terminal operators, shipping carriers, and port basin communities.

**\$2.4M+**

## Annual Terminal Savings

Substantial reduction in avoided demurrage and overtime crane gang allocations.

**-38%**

## Vessel Wait Time

Optimized berth dispatch enables just-in-time pilotage and slow-steaming.

**1.2K tons**

## CO2 Emissions Avoided

Eliminating auxiliary bunker fuel burn while ships idle at outer anchorage basins.

# PortFlow AI: The Future of Smart Ports

Production-ready architecture ready for enterprise cloud scale.

## Production Roadmap

- Live NMEA satellite AIS transponder streaming ingest.
- Direct MongoDB Atlas change-stream triggers.
- Multi-tenant deployment on IBM Cloud / Red Hat OpenShift.

## Team L1 TEAM Y

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- Submission Track: AI Track (IBM Bob Innovation Hackathon)